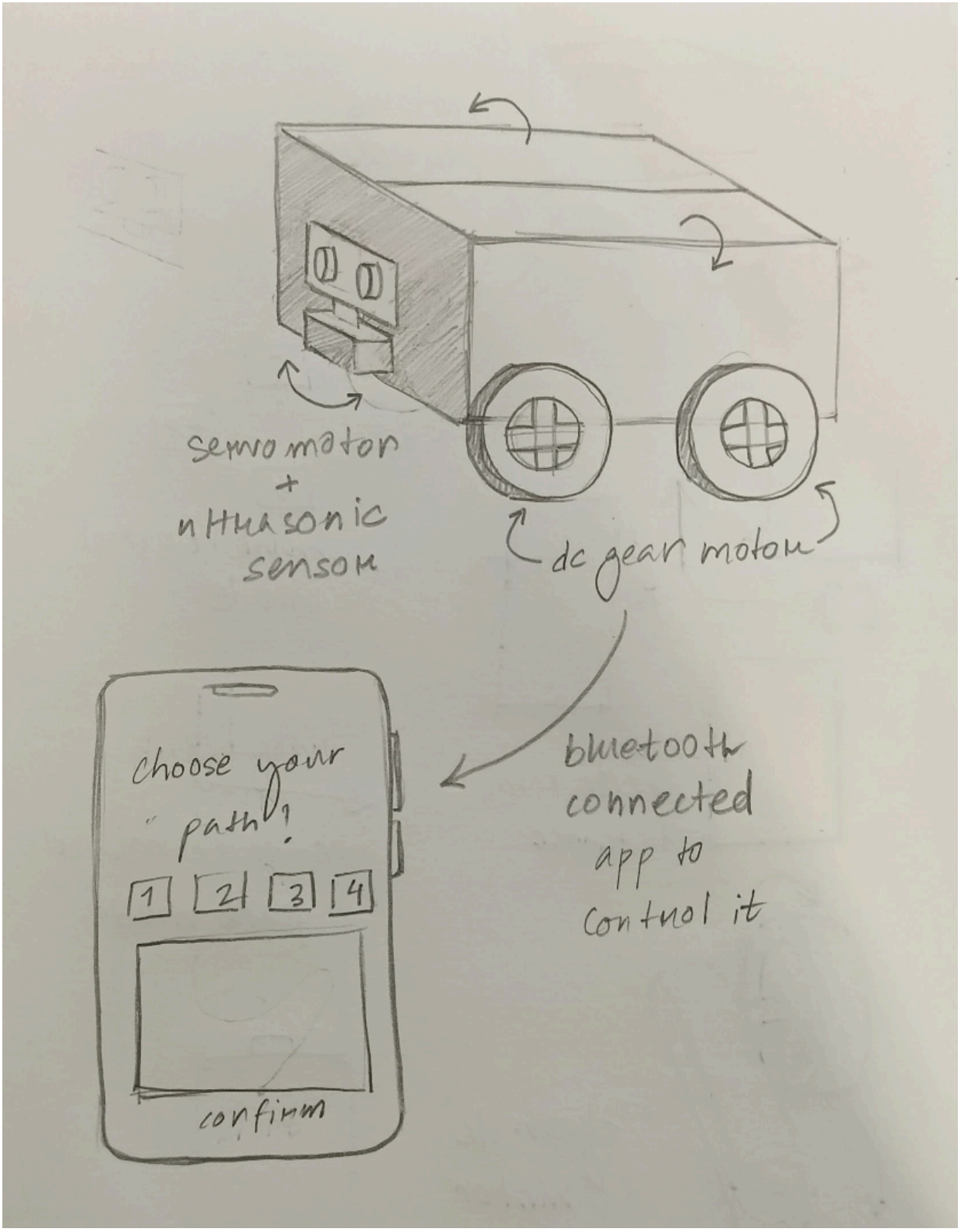
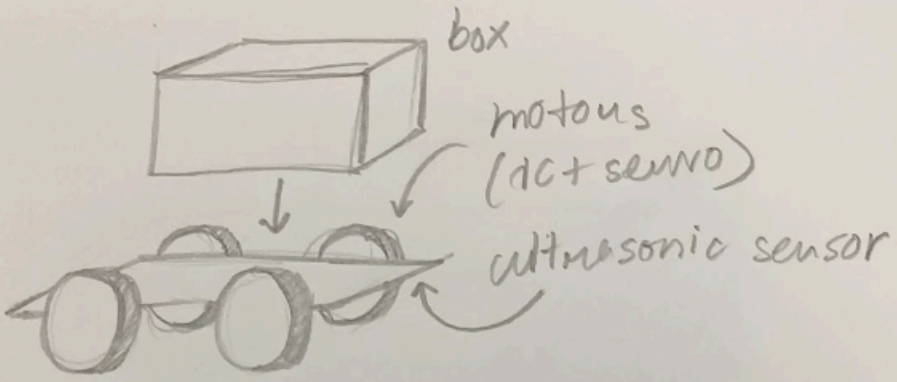
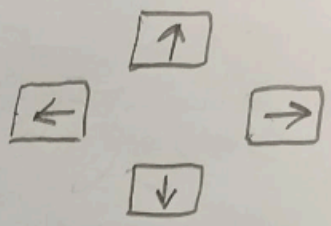
Week 1

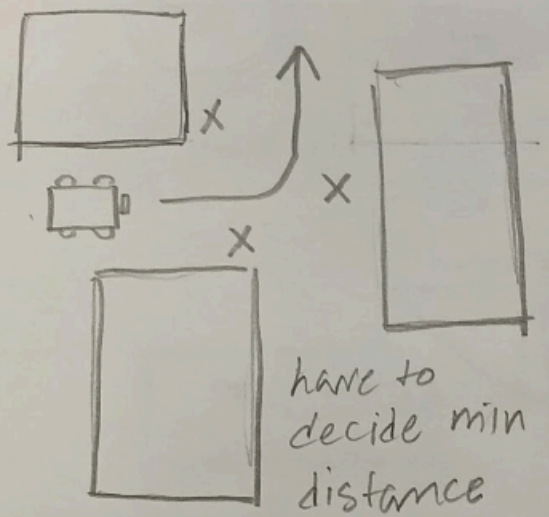
1. Conceptualised and finalised the project idea for the ODT final assignment.
2. Identified the core components such as ultrasonic sensor, DC motors, and servo motor, and mapped out the intended functions of the product.
3. Researched component requirements and sourced all necessary elements.
4. Collaboratively discussed and established the logical flow of the code before beginning development.
5. Developed iterative sketches to visualise and refine the ideal form and design of the product.
6. Finalised the main structure and material selection like laser-cut acrylic sheets for the delivery box and laser-cut MDF for the base, after evaluating the pros and cons of each option.
7. Divided responsibilities across the team, assigning specific code modules and physical fabrication tasks to respective members.
8. Designed a comprehensive circuit diagram to concretise the overall idea and integrate all systems.


servo motor
+
ultrasonic
sensor
dc gear motor
bluetooth
connected
app to
control it
choose your
path!
1
2
3
4
confirm

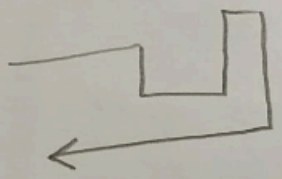




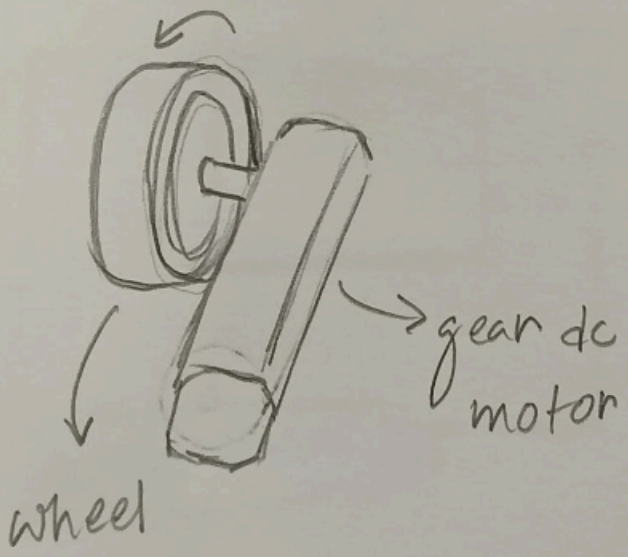
manual control mode

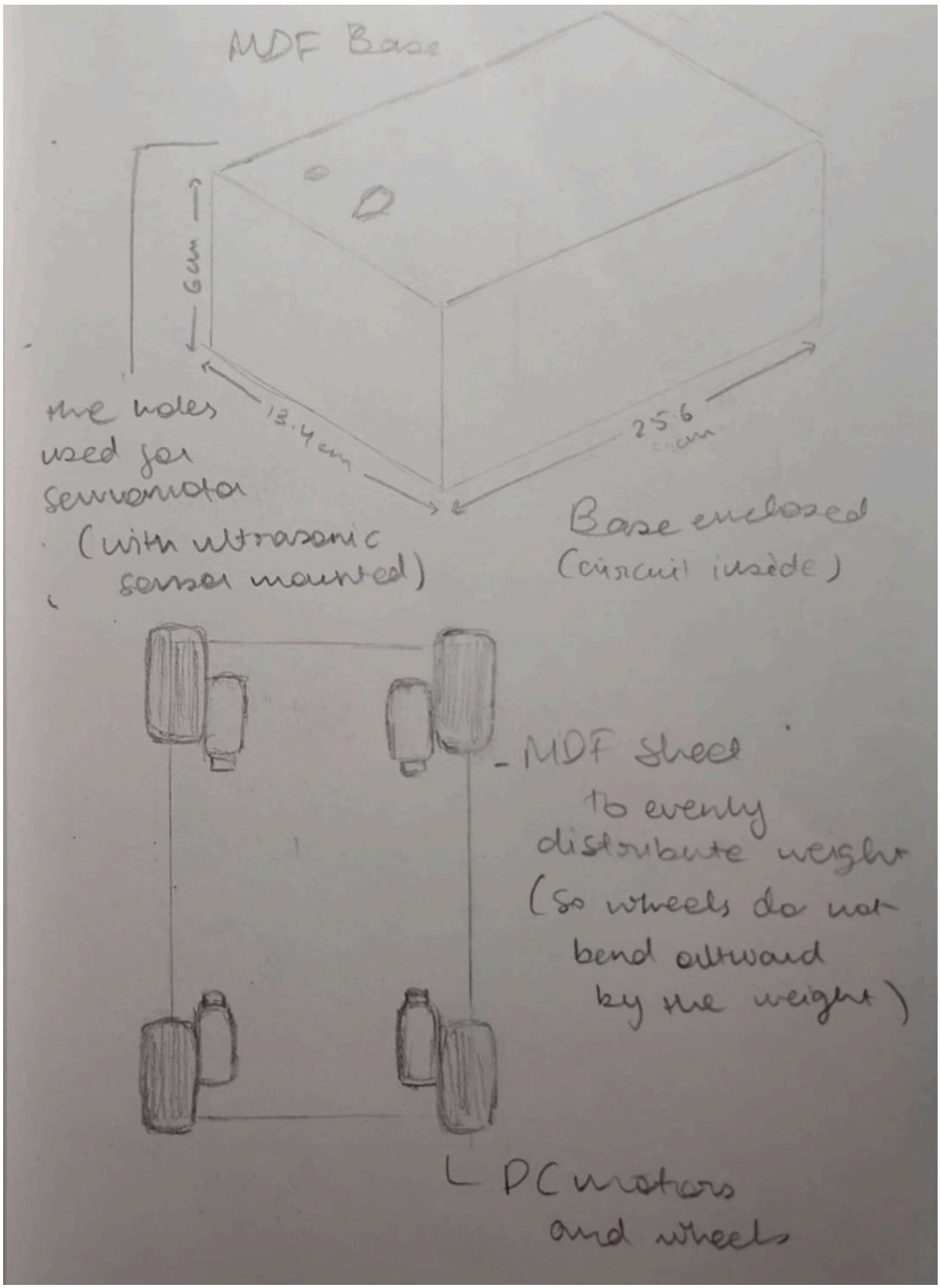


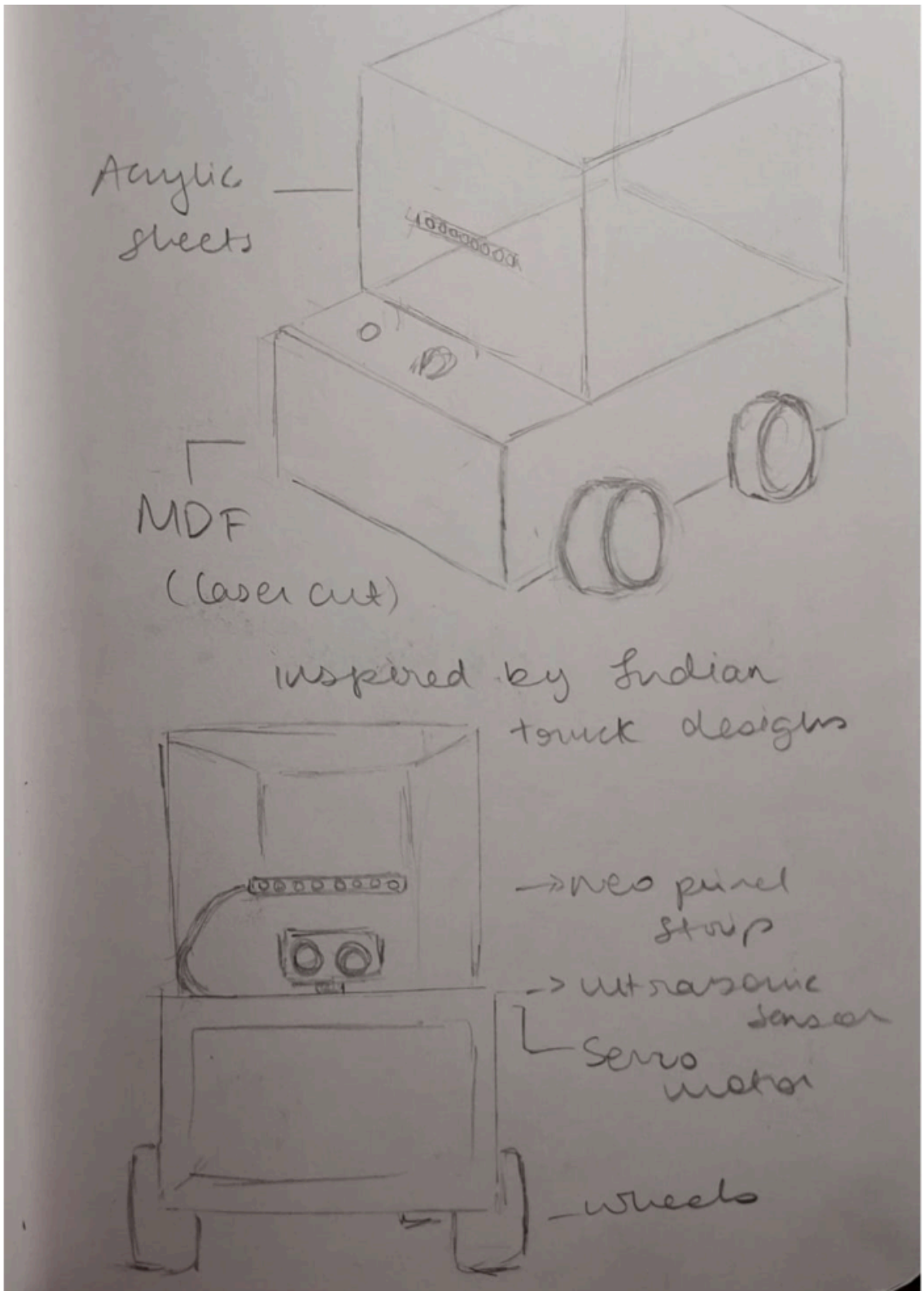
have to decide min distance

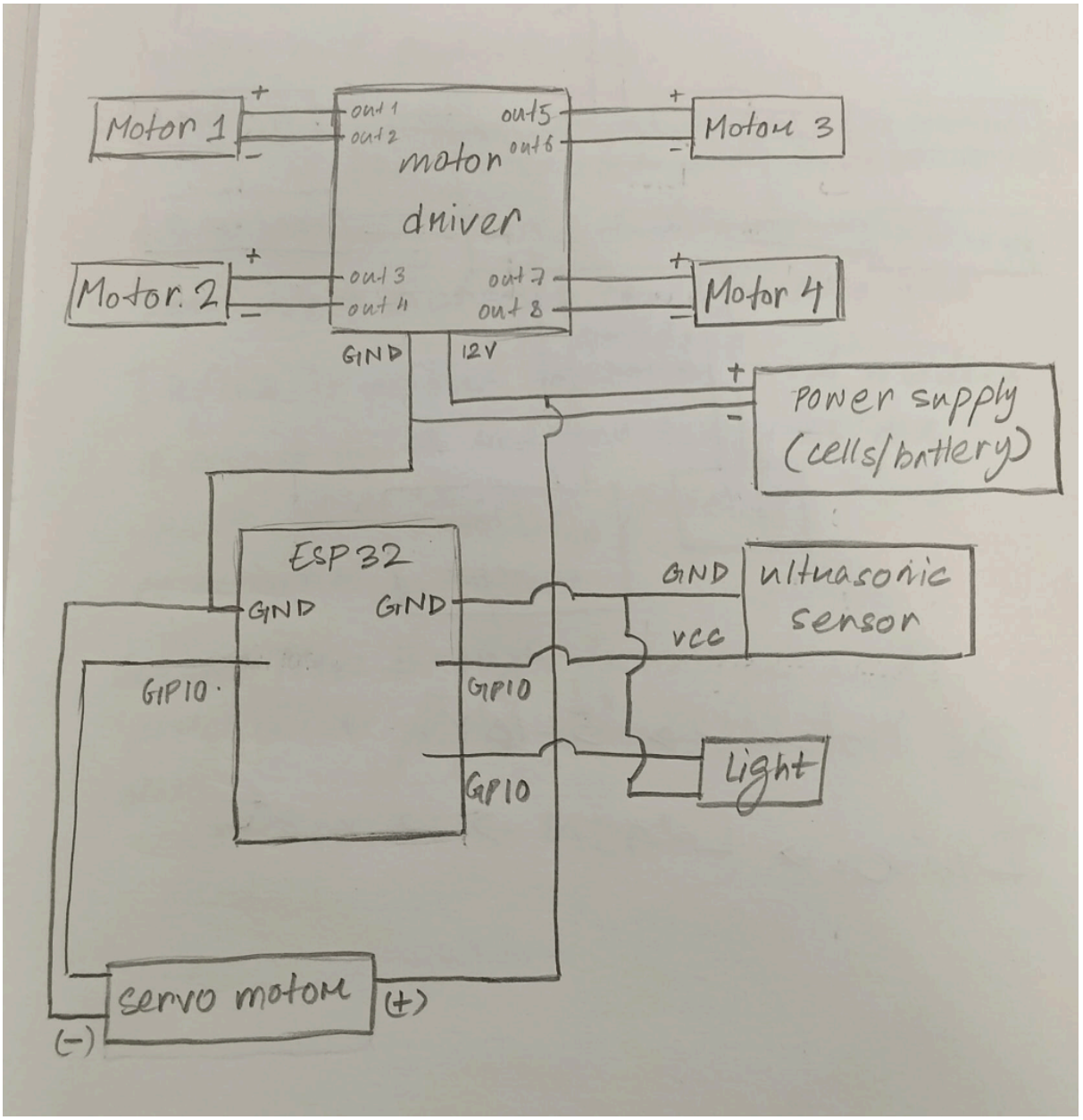


pre-determined path mode




MDF Base
6cm
13.4cm
25.6 cm
the holes used for servomotor (with ultrasonic sensor mounted)
Base enclosed (circuit inside)
MDF sheet to evenly distribute weight (so wheels do not bend outward by the weight)
DC motors and wheels


Acrylic sheets
MDF
(laser cut)
inspired by Indian truck designs
→ neo pixel strip
→ ultrasonic sensor
Servo motor
wheels


Motor 1
Motor 2
Motor 3
Motor 4
out 1
out 2
out 3
out 4
out 5
out 6
out 7
out 8
motor driver
GND
12V
power supply (cells/battery)
ESP32
GND
GND
GPIO
GPIO
GPIO
GND
VCC
ultrasonic sensor
Light
servo motor
(+)
(-)